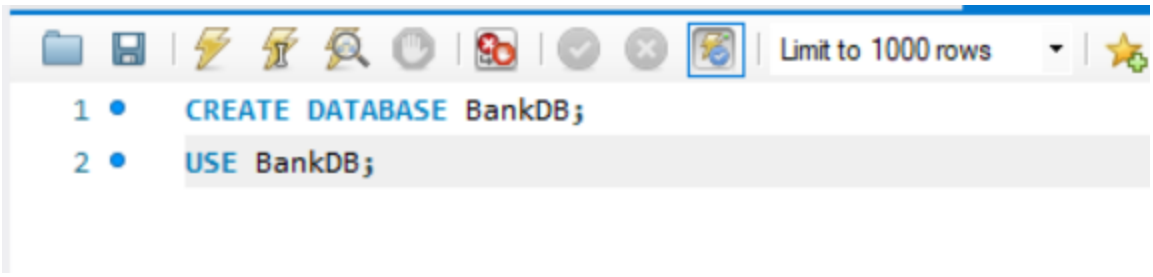
Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

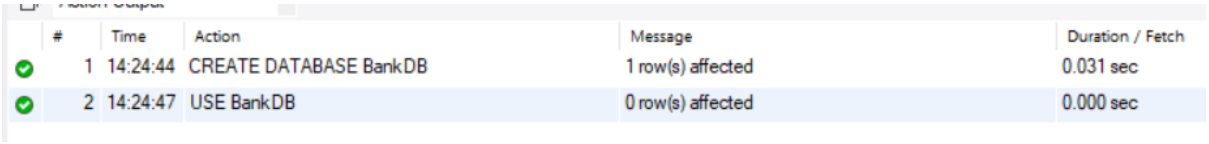
ROLL NO:252003078

NAME: ADAMA SAMARTH

Query 1: -CREATE DATABASE

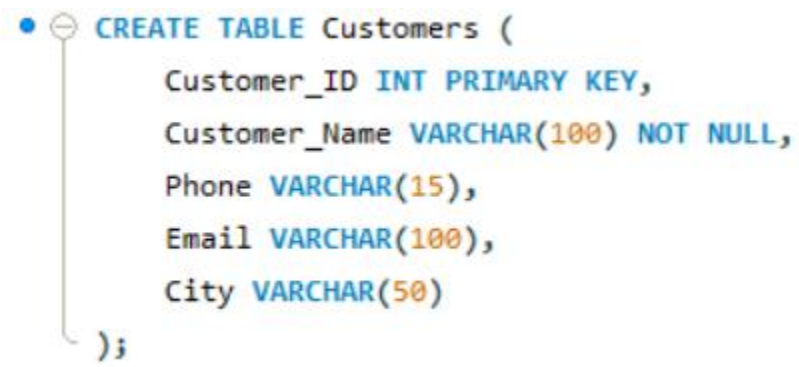


Output: -



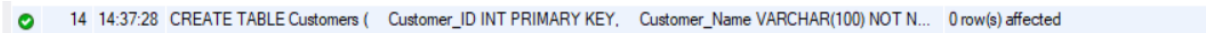
#	Time	Action	Message	Duration / Fetch
1	14:24:44	CREATE DATABASE BankDB	1 row(s) affected	0.031 sec
2	14:24:47	USE BankDB	0 row(s) affected	0.000 sec

Query 2: - CREATE CUSTOMER TABLE



```
CREATE TABLE Customers (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    Email VARCHAR(100),
    City VARCHAR(50)
);
```

Output: -



14	14:37:28	CREATE TABLE Customers (	Customer_ID INT PRIMARY KEY,	Customer_Name VARCHAR(100) NOT N...	0 row(s) affected

Query 3: - CREATE ACCOUNT TABLE

```
);
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2) DEFAULT 0,
    Branch VARCHAR(50),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output: -

```
4  14:28:07  CREATE TABLE Account (  Account_No INT PRI...  0 row(s) affected  0.063 sec
```

Query 4: - CREATE TRANSACTION TABLE

```
);
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);
```

Output: -

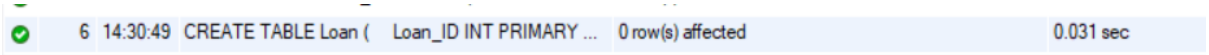
```
5  14:29:36  CREATE TABLE Bank_Transaction (  Transaction...  0 row(s) affected  0.078 sec
```

Query 5: - CREATE LOAN TABLE

```sql
CREATE TABLE Loan (
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output: -



Query 6: - INSERT CUSTOMER DATA

```sql
INSERT INTO Customers
(Customer_ID, Customer_Name, Phone, Email, City)
VALUES
(101, 'Aswitha Patel', '9876543210', 'aswitha@gmail.com', 'Hyderabad'),
(102, 'Adhritha Patel', '9876543211', 'adhritha@gmail.com', 'Vijayawada'),
(103, 'Akshara Reddy', '9876543212', 'akshara@gmail.com', 'Bangalore'),
(104, 'Anirudh Reddy', '9876543213', 'anirudh@gmail.com', 'Chennai'),
(105, 'Ananya Varshitha', '9876543214', 'ananya@gmail.com', 'Hyderabad');
```

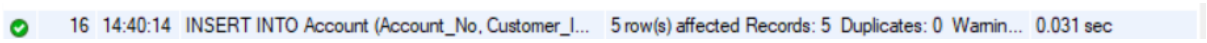
Output: -



Query 7: - INSERT ACCOUNT DATA

```sql
INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 90000, 'Hyderabad');
```

Output: -



Query 8:- INSERT TRANSACTION DATA

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000);
```

Output: -



Query 9: - INSERT LOAN DATA

```
INSERT INTO Loan
(Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
VALUES
(501, 101, 'Home Loan', 5000000, 7.5),
(502, 102, 'Education Loan', 1000000, 6.5),
(503, 103, 'Car Loan', 800000, 8.2),
(504, 104, 'Personal Loan', 500000, 10.5);
```

Output: -



Query 10: - DISPLAY TABLE DATA

```
SELECT * FROM Customers;
```

Output: -

Customer_ID	Customer_Name	Phone	Email	City
101	Aswitha Patel	9876543210	aswitha@gmail.com	Hyderabad
102	Adhritha Patel	9876543211	adhritha@gmail.com	Vijayawada
103	Akshara Reddy	9876543212	akshara@gmail.com	Bangalore
104	Anirudh Reddy	9876543213	anirudh@gmail.com	Chennai
105	Ananya Varshitha	9876543214	ananya@gmail.com	Hyderabad
NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Account;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Bank_Transaction;
```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-15 14:41:48
2	10002	DEPOSIT	15000.00	2026-08-15 14:41:48
3	10003	WITHDRAW	20000.00	2026-08-15 14:41:48
4	10004	DEPOSIT	5000.00	2026-08-15 14:41:48
5	10005	WITHDRAW	10000.00	2026-08-15 14:41:48
NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Loan;
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
NULL	NULL	NULL	NULL	NULL

Query 11: - PROCEDURE TO FIND CUSTOMER ACCOUNTS

```
DELIMITER //
CREATE PROCEDURE GetCustomerAccounts(
    IN p_Customer_ID INT
)
BEGIN
    SELECT
        C.Customer_ID,
        C.Customer_Name,
        A.Account_No,
        A.Account_Type,
        A.Balance,
        A.Balance,
        A.Branch
    FROM Customer C
    JOIN Account A
        ON C.Customer_ID = A.Customer_ID
    WHERE C.Customer_ID = p_Customer_ID;
END //
DELIMITER ;
CALL GetCustomerAccounts(101);
```

Output: -

```
24  14:52:47  CREATE PROCEDURE GetCustomerAccounts(  IN p_Customer_ID INT ) BEGIN  SELECT  C.Custom...  0 row(s) affected
25  14:52:51  CALL GetCustomerAccounts(101)                                                              1 row(s) returned
```

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Aswitha Patel	10001	Savings	50000.00	Hyderabad

Query 12: - PROCEDURE TO DEPOSIT MONEY

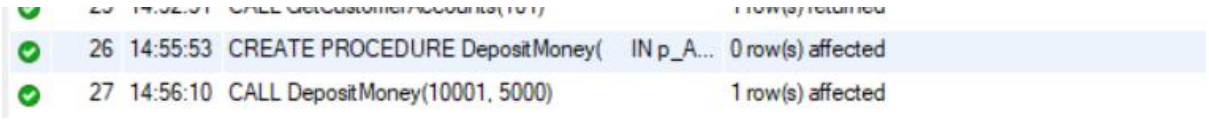
```
DELIMITER //

CREATE PROCEDURE DepositMoney(
    IN p_Account_No INT,
    IN p_Amount DECIMAL(12,2)
)
BEGIN

    UPDATE Account
    SET Balance = Balance + p_Amount
    WHERE Account_No = p_Account_No;

END //
DELIMITER ;
CALL DepositMoney(10001, 5000);
```

Output: -



Query 13: - PROCEDURE TO WITHDRAW MONEY

```
DELIMITER //

CREATE PROCEDURE WithdrawMoney(
    IN p_Account_No INT,
    IN p_Amount DECIMAL(12,2)
)
BEGIN

    UPDATE Account
    SET Balance = Balance - p_Amount
    WHERE Account_No = p_Account_No;

END //

DELIMITER ;
CALL WithdrawMoney(10001, 3000);
```

Output: -

✓	28	14:57:56	CREATE PROCEDURE WithdrawMoney(	IN p...	0 row(s) affected
✓	29	14:58:00	CALL WithdrawMoney(10001, 3000)		1 row(s) affected

Query 14: - TRIGGER – PREVENT NEGATIVE DEPOSIT

```
DELIMITER //
CREATE TRIGGER CheckTransactionAmount
BEFORE INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN

    IF NEW.Amount <= 0 THEN

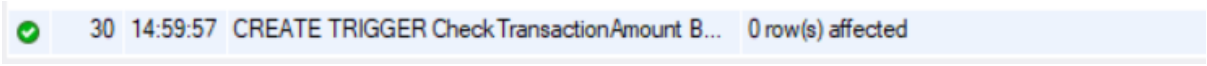
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT =
        'Transaction amount must be greater than zero';

    END IF;

END //

DELIMITER ;
```

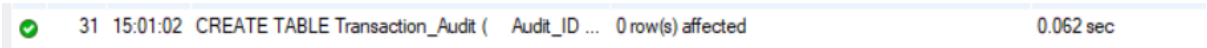
Output: -



Query 15: - CREATE TRANSACTION AUDIT TABLE

```
CREATE TABLE Transaction_Audit (
    Audit_ID INT PRIMARY KEY AUTO_INCREMENT,
    Transaction_ID INT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

Output: -



Query 16: - TRIGGER – TRANSACTION AUDIT

```
DELIMITER //
CREATE TRIGGER TransactionAudit
AFTER INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN

    INSERT INTO Transaction_Audit
    (
        Transaction_ID,
        Account_No,
        Transaction_Type,
        Amount
    )
    VALUES
    (
        NEW.Transaction_ID,
        NEW.Account_No,
        NEW.Transaction_Type,
        NEW.Amount
    );

END //

DELIMITER ;
```

Output: -

```
31  15:01:02  CREATE TABLE Transaction_Audit (  Audit_ID ...  0 row(s) affected  0.062 sec
```

Test :-

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 2500);
```

Check: -

```
32  15:04:10  INSERT INTO Bank_Transaction (Account_No, ...  1 row(s) affected  0.031 sec
```

Query 17: - PROCEDURE – ACCOUNT TRANSFER

```
DELIMITER //
CREATE PROCEDURE TransferMoney(
    IN SenderAccount INT,
    IN ReceiverAccount INT,
    IN TransferAmount DECIMAL(12,2)
)
BEGIN

    DECLARE SenderBalance DECIMAL(12,2);

    SELECT Balance
    INTO SenderBalance
    FROM Account
    WHERE Account_No = SenderAccount;

    IF SenderBalance < TransferAmount THEN

        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT =
        'Transfer failed: Insufficient balance';
```

```
        'Transfer failed: Insufficient Balance';

    ELSE

        UPDATE Account
        SET Balance = Balance - TransferAmount
        WHERE Account_No = SenderAccount;

        UPDATE Account
        SET Balance = Balance + TransferAmount
        WHERE Account_No = ReceiverAccount;

    END IF;

END //
```

```
DELIMITER ;
CALL TransferMoney(10001, 10002, 5000);
```

Output: -

✔	34	15:08:19	CREATE PROCEDURE TransferMoney(	IN Se...	0 row(s) affected	0.04
✔	35	15:08:53	CALL TransferMoney(10001, 10002, 5000)		1 row(s) affected	0.031 sec

CHECK

```
SELECT *
FROM Account
WHERE Account_No IN (10001,10002);
```

Output: -

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	47000.00	Hyderabad
	10002	102	Savings	80000.00	Vijayawada
*	NULL	NULL	NULL	NULL	NULL

Query 18: - PROCEDURE WITH IN, OUT PARAMETERS

```
DELIMITER //

CREATE PROCEDURE GetBalance(
    IN p_Account_No INT,
    OUT p_Balance DECIMAL(12,2)
)
BEGIN

    SELECT Balance
    INTO p_Balance
    FROM Account
    WHERE Account_No = p_Account_No;

END //

DELIMITER ;
CALL GetBalance(10001, @CurrentBalance);
SELECT @CurrentBalance;
```

Output: -

✓	37	15:11:44	CREATE PROCEDURE GetBalance(IN p_Acc...	0 row(s) affected
✓	38	15:11:48	CALL GetBalance(10001, @CurrentBalance)	1 row(s) affected

Result Grid | Filter R

	@CurrentBalance
▶	47000.00

Query 19: -TRIGGER COMMANDS

```
SHOW TRIGGERS;
```

Output: -

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	Trigger	Event	Table	Statement	Timing	Created	sql_mode	De
▶	CheckTransactionAmount	INSERT	bank_transaction	BEGIN IF NEW.Amount <= 0 THEN SIG...	BEFORE	2026-08-15 14:59:57.20	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE...	roo

Query 20: - PROCEDURE MANAGEMENT COMMANDS

```
SHOW PROCEDURE STATUS
WHERE Db = 'BankDB';
DROP PROCEDURE GetAllCustomers;
```

Output: -

	Db	Name	Type	Definer	Modified	Created	Security_type	Comment	character_set_client	collation_connection	Databas Collation
▶	bankdb	DepositMoney	PROCEDURE	root@localhost	2026-08-15 14:55:53	2026-08-15 14:55:53	DEFINER		utf8mb4	utf8mb4_0900_ai_ci	utf8mb4
	bankdb	GetBalance	PROCEDURE	root@localhost	2026-08-15 15:11:44	2026-08-15 15:11:44	DEFINER		utf8mb4	utf8mb4_0900_ai_ci	utf8mb4
	bankdb	GetCustomerAccounts	PROCEDURE	root@localhost	2026-08-15 14:52:47	2026-08-15 14:52:47	DEFINER		utf8mb4	utf8mb4_0900_ai_ci	utf8mb4
	bankdb	TransferMoney	PROCEDURE	root@localhost	2026-08-15 15:08:19	2026-08-15 15:08:19	DEFINER		utf8mb4	utf8mb4_0900_ai_ci	utf8mb4
	bankdb	WithdrawMoney	PROCEDURE	root@localhost	2026-08-15 14:57:56	2026-08-15 14:57:56	DEFINER		utf8mb4	utf8mb4_0900_ai_ci	utf8mb4